Quantifying the Cost of Allowing Major League Baseball Players to Call Pitches

Andersen Pickard

Rice University

SMGT 490

Dr. Hua Gong

May 3, 2026

Abstract

Pitch selection in Major League Baseball has remained a largely intuitive process, historically delegated to the catcher despite baseball's broader embrace of data analytics. This paper proposes and implements a machine learning framework to quantify the cost of suboptimal pitch selection decisions and evaluate the performance of pitchers, catchers, teams, and pitcher-catcher batteries. Using Statcast pitch-level data from the 2021 through 2025 MLB seasons, a LightGBM gradient boosting model is trained to predict runs lost as a function of game situation and pitch characteristics. The trained model is then applied to generate counterfactual pitch scenarios for every pitch thrown in 2025, estimating the expected run value of each pitch type a pitcher could have thrown in a given situation. The pitch type with the lowest predicted run value is identified as the optimal selection, and the difference between the predicted run value of the actual pitch and the optimal pitch is quantified as runs lost. Aggregated across pitchers, catchers, teams, and batteries, these pitch-level estimates reveal substantial variation in pitch selection quality across the league. League-wide, MLB players selected the optimal pitch type just 23.1% of the time in 2025. The findings suggest that predictive modeling tools capable of transmitting optimal pitch calls from the dugout to the field in real time could offer organizations a meaningful and largely untapped competitive advantage over their opponents, resulting in potentially 100+ additional runs per season.

Introduction

Baseball started as a simple game but has gradually developed into one of the most complex, thought-provoking sports amid the rise of data analytics. In the decades since the release of Michael Lewis' *Moneyball* (2003), Major League Baseball organizations have clamored to gain marginal — sometimes even negligible — advantages over their opponents by evaluating data, implementing statistical methods and models, and refining traditional processes.

Almost no aspect of baseball has gone unchanged amid the rise of data analytics. Scouting departments are becoming more metric-based, baseball operations departments are introducing highly advanced and proprietary models, player development staffs are integrating biomechanical practices at a larger rate than ever, and coaching staffs are launching exhaustive data processes to gain even a fraction of a run in a series against their upcoming opponent.

Even aspects of baseball that are independent from the 30 MLB organizations have been influenced by this data evolution. Agents use intensive data processes to quantify a client's worth, television networks run simulations to decide which games to air in which regional market and time slot, sportsbooks run intensive models nonstop to stay a step ahead of their bettors, and even players themselves dive deeper into their own data through independent training organizations.

While the data revolution has launched sweeping changes across the sport of baseball, one area has remained quite traditional. Pitch calling, which is the process of selecting a certain pitch type to throw in a certain scenario, has incorporated basic data scouting but currently lags behind in terms of predictive models.

As has been the case for decades, pitchers place trust in their daily catcher to call pitches. When a batter comes up to the plate, a thorough analysis of data is abandoned in favor of a somewhat arbitrary process left up to the decision-making skills of the catcher. Sometimes, this catcher has no previous experience facing the specific batter, catching for the specific pitcher, or even catching in the majors entirely.

In a world where coaches and data experts have their fingerprints on every baseball strategy decision, pitch calling remains a largely arbitrary process placed upon the catcher's gut feeling under a short window of time.

Herein lies a problem in how organizations are advancing throughout games, as well as a ripe opportunity for these organizations to embrace data, maximize change in run and win expectancy, and model outcomes to gain a significant competitive advantage.

This paper proposes a process for MLB organizations to use predictive modeling to relay the ideal pitch type from the dugout to the catcher in real time. Teams no longer have to rely on a catcher who hasn't been able to review scouting reports for several hours, and instead can use real-time predictive modeling tools in the dugout to transmit pitch calls from a coaching staff member directly to their pitcher.

It's worth noting that there is precedent for a process that shifts pitch-calling responsibility from catchers to coaches. For an undisclosed duration of games in September of the 2025 MLB season, the Miami Marlins transmitted pitch calls from a coach in the dugout to the players on the field, including the pitcher and catcher. It's unclear whether the Marlins were using a model tool — like the one described in this paper — to call pitches or if a coach was simply making suggestions similar to how a catcher would. However, the bottom line is that the Marlins showed that pitch call transmission from the dugout to the field is not only permitted under MLB rules, but also possible.

This paper proposes that MLB organizations introduce predictive modeling to calculate the expected run value of a pitcher's possible pitch types, given their existing arsenal, pitch shapes and metrics, and past results. The statistical breadth of this research includes comparing "real" pitches (pitches actually thrown in MLB situations from 2021–2025) to "theoretical" pitches (alternative pitches that could have been thrown in those same situations). An expected run value is attributed to both the real and theoretical pitch type, and the pitch type with the lowest expected run value is treated as the "ideal" selection. If the real and ideal pitch types are the same, then the catcher will be credited for making the best possible pitch call. Otherwise, we will acknowledge that the catcher could have made an alternative pitch call to improve run and win expectancy. The effect of pitch calling can be quantified not only to evaluate individual catchers, but pitchers and teams, too.

The modeling process produces quantifiable metrics, leaderboards, and visualizations to see which players and teams excelled most and least in terms of pitch calling. However, the overall results can be used generally to justify why or why not teams should begin calling pitches from the dugout.

Literature Review

Significant research into the effects and strategies of pitch sequencing currently exists. As more data becomes available throughout baseball, both internally and publicly, additional research has been conducted to evaluate pitch sequencing, its value and effectiveness, and the ideal approach to selecting which pitch to throw. Advancements in this space have been assisted by various developments across Major League Baseball, including the implementation of HawkEye cameras within the last decade and the recent availability of bat-tracking data.

Bock (2015) examines whether machine-learning models can predict a pitcher's next pitch in real time and place a quantifiable value behind pitchers' long-term performances. Using pitch-by-pitch data from 2011 through 2013, Bock constructs models for four pitch types and incorporates variables such as pitch history, count, base state, and handedness. The author uses a predictability index to evaluate the accuracy of their model across pitch types and to inform the prediction for the next pitch type. The average accuracy of their model's ability to predict the next pitch type was 74.5%. This research improved upon prior work in the field by introducing a more complex pitch classification system; it identified four pitch types instead of just two (fastball and non-fastball). It also exceeded the accuracy scores of prior work in the space.

The results of Bock's research indicate that pitch selection is somewhat predictable, yet also varies across game situations. Predictability was higher when batters had the handedness advantage, and predictability varied significantly based on count.

Interestingly, Bock finds a statistically significant relationship between pitch predictability and long-term performance metrics for the middle range of predictability values. Their application of linear regression models shows that increased predictability is associated with higher ERA and FIP, which are indicators of poor pitcher performance. With a small R-squared value, though, these relationships explain only a small portion of the variance in ERA and FIP, and that when looking at just the most elite class of pitchers, they perform at a high level (lower FIP and ERA) despite their pitch choices being more predictable. With more analysis, future research could examine whether "ideal" pitch sequencing decisions offer a greater positive change in performance metrics for weaker-performing pitchers (higher FIP and ERA) and a positive but lesser change in performance metrics for stronger-performing pitchers (lower FIP and ERA).

Prasad (2021) explores how pitchers construct their own pitch sequences over a period of time. The author emphasizes that pitchers and batters develop a "memory" of pitch types, either helping or hurting their ability to perform at the highest level, while traditional models that focus on pitch sequencing fail to have any sort of memory. Armed with pitch-by-pitch data from 2015 through 2019, the author uses directed graphs to encode the sequence of pitch types that a pitcher follows. This allows for the development of recurring patterns that go beyond back-to-back pitch sequencing and instead cover a larger timeframe of pitches thrown; or, in other words, develops a sort of memory. The main goal of the author is to investigate whether pitch sequencing tendencies are distinct from the makeup of a pitcher's arsenal.

The results of Prasad's research provide evidence that recurring patterns can be observed in pitch sequences across both pitch type and pitch location. By interpreting the directed graphs, Prasad identifies components of pitch sequencing defined as "setup" and "knockout" pitches, making these classifications based on leverage. The author finds that pitchers have consistent sequencing tendencies over entire seasons, suggesting that pitch sequencing strategy is a stable component of a pitcher's long-term behavior.

Prasad's research also shows that pitch sequencing patterns are not entirely determined by pitch arsenal alone. Even pitchers who have similar pitch mixes will have different strategies for pitch sequencing, which indicates that pitch sequencing decisions are independent and unique to each individual pitcher.

While Bock and Prasad focus on empirical regularities in pitch sequencing, Melville (2023) applies a game theory model to the concept of pitch sequencing and the matchup between a pitcher and batter. They model pitch selection as a process that requires pitchers to weigh unpredictability and pitch effectiveness, and requires batters to weigh deviations from optimal randomness. Using historical pitch-outcome data, the paper identifies certain pitch-mix strategies that specify optimal sequencing probabilities based on different count and matchup scenarios. The author's goal is to determine whether the pitch sequencing strategy imposed by their model outperforms actual, historical data and examples.

The results from Melville's research suggest that many pitchers deviate from optimal pitch mixes. Not only do they overuse certain pitches, but they have a tendency to overuse those pitches in repeated, predictable counts, such as pitcher-ahead or two-strike scenarios. The paper suggests that observing the optimal pitch mix proposed by the game theory model would reduce the expected contribution from the opposing batter, and it would have a greater effect on some batters more than others. The analysis also shows that optimal strategies vary substantially across pitch arsenals, implying that the quantitative effectiveness of pitch sequencing is different for each pitcher depending on the pitches they throw and the characteristics of those pitches.

Morishita et al. (2025) investigate the relationship between pitch selection (from a batter's perspective), executive function, and batting performance in college baseball. Pitch selection is defined as a batter's ability to swing at would-be strikes that produce optimal outcomes, and avoid would-be balls or any would-be strikes that would be difficult to hit. The study recruited 14 Division I baseball players to complete a virtual reality hitting and pitch selection task, with results showing a correlation between pitch selection and on-base percentages, as well as a correlation between pitch selection and walk rates. While this study focuses on batter pitch selection and not pitcher pitch sequencing, it provides a framework for quantifying the value of making good and bad pitch selections and swing decisions from a batter's perspective. This framework suggests that a batter gains value from making the correct pitch selection and swing decisions, just like a pitcher gains value from making an ideal pitch call or sequencing decision.

These findings from Morishita et al. suggest that poor pitch sequencing decisions could create an advantage for the hitter, even when pitch-level characteristics (i.e., shape) and metrics (i.e., Stuff+) remain ideal, because the poor pitch sequencing decisions make it easier for a batter to identify and damage the incoming pitch.

Bock's research opens the door for the potential to place an empirical value on ideal and non-ideal pitch sequencing decisions, especially since there are indications that "ideal" pitch sequencing decisions offer a greater positive change in performance metrics for weaker-performing pitchers than stronger-performing pitchers. However, the study still has some limitations. The data excludes pitchers with fewer than 1,000 pitches, limiting eligibility to a select, often "better" group of pitchers. Meanwhile, it ignores many relievers, as well as starters who made around 10 to 12 starts or fewer, whether due to performance, roster, or injury reasons. Specifying a minimum sample size threshold could still be necessary, but leaving that threshold as high as 1,000 pitches drastically changes which types of pitchers will be analyzed and also changes potential conclusions that can be drawn from the investigation. One could argue that pitchers with fewer than 1,000 pitches are exactly the population that would benefit most from tools to identify their ideal pitch sequencing tendencies, especially if the theory that pitch sequencing impacts weaker-performing pitchers more than stronger-performing pitchers holds true.

Meanwhile, Melville's model does not incorporate pitch location or the tunneling effects of pitch sequencing, and it could be expanded to incorporate more advanced ball- and pitch-tracking techniques. Finally, Prasad's study also leaves room for further advancement, as the paper stops short of attributing changes in outcome-based performance metrics (run value, whiff rates, xWOBA, etc.) to pitch sequencing decisions.

Methodology

This study employs a machine learning framework to evaluate pitch selection decisions in Major League Baseball, using Statcast pitch-level data from the 2021 through 2025 MLB seasons. The objective of the study is to estimate the run value of any pitch type a player could have thrown in a given situation, and then compare that against the estimated run value of what pitch was actually thrown in that situation. This process quantifies the cost (the net runs lost) of a suboptimal pitch being selected, rather than the optimal pitch type.

Pitch-level Statcast data was collected for all MLB games from 2021 through 2025 by scraping from Baseball Savant. Several physical measurements were converted to inches to ensure consistency across variables, including pitch movement, release extension, and release position coordinates. The sign (positive or negative) on horizontal movement was flipped to better reflect widespread baseball practices. The updated horizontal movement metrics can be interpreted from the pitcher's perspective looking toward home plate, with negative horizontal break indicating gloveside movement for a right-handed pitcher and armside movement for a left-handed pitcher.

The outcome variable, runs scored, was derived by computing the difference between the batting team's score before and after each pitch. For the sake of consistency and to better adhere with the spirit of conducting original research, the model makes new predictions on the runs scored variable, rather than using the existing delta run expectancy variable from the Statcast data.

For each pitcher-season combination (one row per unique pitcher per unique season), average metrics were computed across each pitch type in a pitcher's arsenal. Metrics include velocity, horizontal and induced vertical break, spin rate, spin axis, release extension, release position, and velocity and acceleration components. It should be noted that although these metrics were calculated for the purpose of summarizing data, not all pitch type variables were used in the model training phase, nor were they all used to construct counterfactual pitch scenarios.

A gradient boosting model was trained using LightGBM on all pitch events from 2021–2024, with runs scored as the target variable. The selected features capture the situational context of each pitch (row), including but not limited to count, outs, inning, runner state, and score, as well as physical pitch features, including but not limited to release speed, horizontal and induced vertical break, spin rate, spin axis, and release extension. Batter and pitcher handedness were also included for platoon effects. The model effectively learns the expected run-scoring outcome (expected run value) associated with any possible combination of game situation and pitch characteristics.

The trained model was applied to all 2025 pitch-by-pitch events to evaluate pitch selection decisions. In an effort to remove "position players pitching" from the dataset, rows were dropped if the pitcher had 50+ plate appearances as a batter in 2025 and was not named Shohei Ohtani. For each pitch in the test set, a candidate set was constructed by joining the actual pitch event to every other possible pitch type in that pitcher's 2025 arsenal. For example, if a pitcher was capable of throwing a sinker, slider, and changeup, and they threw a changeup in a certain scenario, then counterfactual pitches would be generated for a sinker and slider in that example.

Producing these counterfactual pitches results in one candidate row per pitch type, each representing the hypothetical scenario of throwing that pitch type in the exact same game situation as the actual pitch that was thrown.

The counterfactual pitches are generated from the pitcher's season mean metrics for that pitch type. For the actual pitches that were thrown, the empirical data is preserved from the actual pitch event; no new assumptions are made and no new means are applied.

The model then generated a predicted run value for each candidate row. For each original pitch event, the candidate with the lowest predicted run value was identified as the optimal pitch recommendation, representing the pitch type that the model indicates should have been thrown to minimize run expectancy in that situation.

Decision quality was evaluated by comparing the model's predicted run value for counterfactual pitches with the actual pitch that was thrown. The difference between those two values, referred to as runs lost, represents the estimated run expectancy cost of the pitch selection decision on each pitch. If the ideal pitch was thrown, the runs lost value is 0, because the model indicates the best possible pitch was thrown and no changes should have been made to the pitch selection. If a counterfactual pitch should have been thrown instead of the actual pitch, then the runs lost value is greater than 0, because the model indicates the pitch selection decision cost a team runs.

These pitch-level estimates were then aggregated at four different levels — pitcher, catcher, team, and battery — to produce season-level evaluations of pitch selection quality, indicating runs lost over the season, runs lost per pitch, and proportion of pitches on which the optimal pitch type was selected. ("Battery" refers to a pitcher-catcher pairing; the battery is identified by the pitcher name plus catcher name.)

Results

This research revealed noteworthy cases of pitchers, catchers, teams, and batteries losing substantial amounts of runs due to pitch selection decisions. Some pitchers, catchers, teams, and batteries fared better than others, leaving the door open for organizations to potentially respond with strategical changes on a case-by-case basis.

Pitcher Jack Little (Los Angeles Dodgers) had the most runs lost on a per-pitch basis in 2025, at 0.0338 runs per pitch. He lost 1.1814 runs over the course of the full season, and he and his catcher selected the optimal pitch just 17.14% of the time.

Table 1: Pitchers losing the most runs per pitch

Player	Optimal %	Runs Lost	Per Pitch
Jack Little	0.1714	1.1814	0.0338
Cole Wilcox	0.2564	1.2990	0.0333
Philip Abner	0.4337	2.2493	0.0271

Pitcher Garrett Acton (Tampa Bay Rays) had the fewest runs lost on a per-pitch basis in 2025, at 0.0004 runs per pitch. He lost just 0.008 runs over the course of the full season, and he and his catcher selected the optimal pitch 80.95% of the time.

Table 2: Pitchers losing the fewest runs per pitch

Player	Optimal %	Runs Lost	Per Pitch
Garrett Acton	0.8095	0.0080	0.0004
Ryan Burr	0.6410	0.0240	0.0006
Julian Fernández	0.5000	0.0489	0.0006

Catcher Anthony Seigler (Milwaukee Brewers) had the most runs lost on a per-pitch basis in 2025, at 0.0165 runs per pitch. He lost 0.5615 runs over the course of the full season, and he and his pitcher selected the optimal pitch just 17.65% of the time.

Table 3: Catchers losing the most runs per pitch

Player	Optimal %	Runs Lost	Per Pitch
Anthony Seigler	0.1765	0.5615	0.0165
Blake Sabol	0.2419	9.8565	0.0153
Garrett Stubbs	0.3048	1.2463	0.0119

Catcher Chad Wallach (Los Angeles Angels) had the fewest runs lost on a per-pitch basis in 2025, at 0.0005 runs per pitch. He lost just 0.0221 runs over the course of the full season, and he and his pitcher selected the optimal pitch 53.33% of the time.

Table 4: Catchers losing the fewest runs per pitch

Player	Optimal %	Runs Lost	Per Pitch
Chad Wallach	0.5333	0.0221	0.0005
Moisés Ballesteros	0.2907	0.1805	0.0021
Rafael Flores	0.2378	0.4049	0.0028

The battery of Sam Long and Luke Maile (Kansas City Royals) had the most runs lost on a per-pitch basis in 2025, at 0.0325 runs per pitch. They lost 1.7551 runs over the course of the full season, and together they selected the optimal pitch just 9.26% of the time.

Table 5: Batteries losing the most runs per pitch

Battery	Pitches	Optimal %	Runs Lost	Per Pitch
Sam Long & Luke Maile	54	0.0926	1.7551	0.0325
José Castillo & Adley Rutschman	97	0.2062	3.0169	0.0311
Philip Abner & James McCann	61	0.4426	1.8223	0.0299

The battery of Will Vest and Tomás Nido (Detroit Tigers) had the fewest runs lost on a per-pitch basis in 2025, at 0.0006 runs per pitch. They lost just 0.0209 runs over the course of the full season, and together they selected the optimal pitch 41.67% of the time.

Table 6: Batteries losing the fewest runs per pitch

Battery	Pitches	Optimal %	Runs Lost	Per Pitch
Will Vest & Tomás Nido	36	0.4167	0.0209	0.0006
Mason Miller & Jhonny Pereda	37	0.4865	0.0235	0.0006
Kenley Jansen & Sebastián Rivero	28	0.3929	0.0182	0.0006

A fifth-inning pitch from Sam Aldegherri (Athletics) to Jake Burger (Rangers) on July 8, 2025, resulted in the greatest pitch selection-based run expectancy loss (0.6764 runs).

Table 7: Pitches with the greatest run expectancy loss

Date	Inning	Pitcher	Catcher	Batter	Runs Lost	Pitch	Description
2025-07-08	5	Sam Aldegheri	Logan O'Hoppe	Jake Burger	0.6764	SL	hit_into_play
2025-08-02	7	Danny Coulombe	Kyle Higashioka	Cal Raleigh	0.6221	FC	foul
2025-08-02	7	Danny Coulombe	Kyle Higashioka	Cal Raleigh	0.6011	FC	foul
2025-07-06	7	Greg Weissert	Connor Wong	Luis García	0.5503	FF	foul
2025-08-21	7	Danny Coulombe	Jonah Heim	Luke Maile	0.5430	FC	foul

The Boston Red Sox had the most runs lost on a per-pitch basis in 2025, at 0.0085 runs per pitch. They lost 188.967 runs over the course of the full season, and they selected the optimal pitch just 19.69% of the time.

Table 8: Teams losing the most runs per pitch

Team	Optimal %	Runs Lost	Per Pitch
BOS	0.1969	188.9670	0.0085
STL	0.2165	178.2737	0.0082
WSH	0.2343	187.2721	0.0082

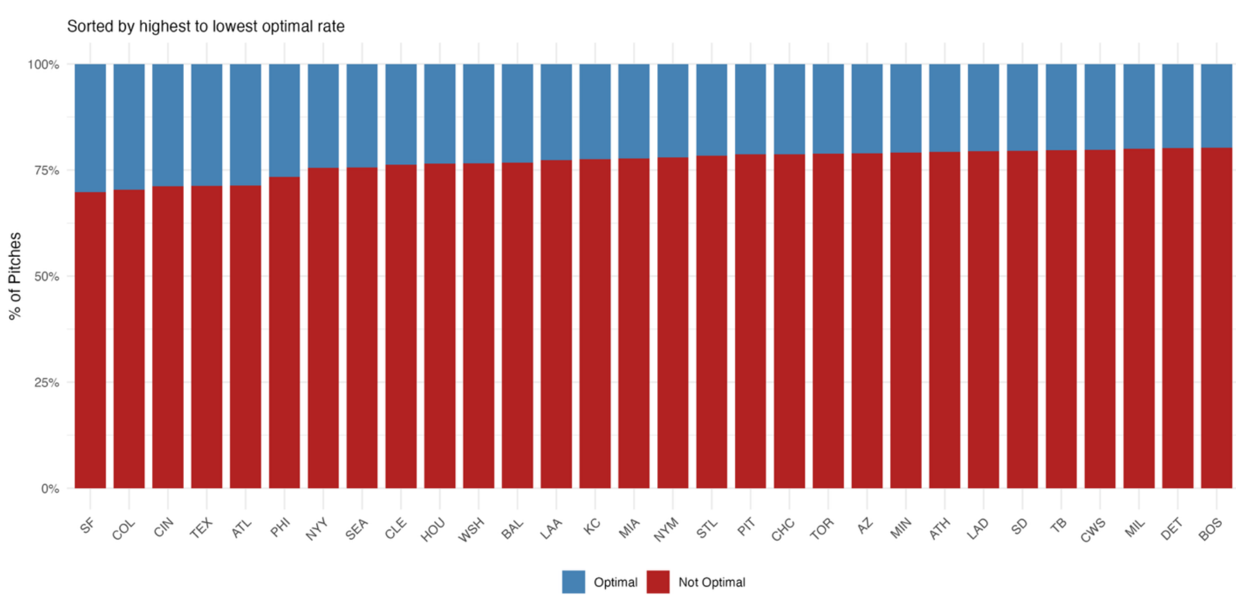
The Chicago Cubs had the fewest runs lost on a per-pitch basis in 2025, at 0.0059 runs per pitch. They lost just 124.8151 runs over the course of the full season, and they selected the optimal pitch 21.3% of the time.

Table 9: Teams losing the fewest runs per pitch

Team	Optimal %	Runs Lost	Per Pitch
CHC	0.2130	124.8151	0.0059
ATL	0.2867	130.7860	0.0059
TEX	0.2872	126.8987	0.0060

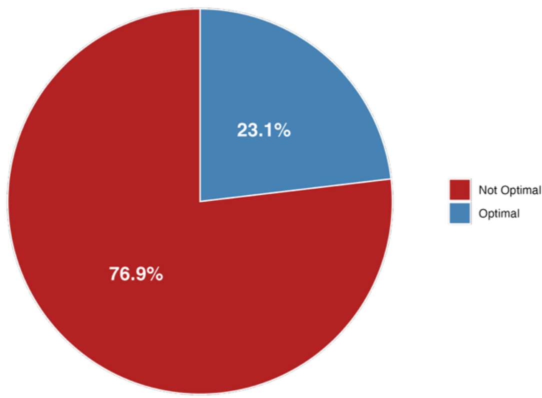
The following graph examines the above idea of team-level decisions in a slightly different manner. Whereas the tables above are sorted by runs lost per pitch, the following graph shows a breakdown of optimal versus non-optimal pitch rates, sorted from most optimal to least optimal. While the Red Sox remain the least successful, we can observe some variations between the run cost totals and the optimal proportions. For example, the Cubs had the second-fewest runs lost per pitch, but the 12th-worst ratio of optimal to non-optimal pitches. A result like this indicates that the effect of the Cubs' pitch selection decisions was far more extreme in beneficial scenarios; when they made the right pitch selection, the expected run lost totals were abnormally small, and when they made the wrong pitch selection, the expected run lost totals were not too large.

Graph 1: Optimal pitch selection rates by team



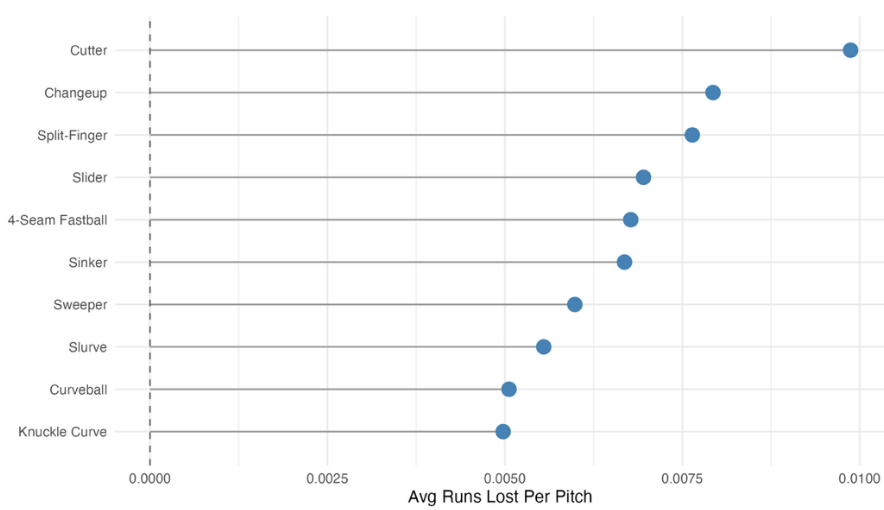
The following graph shows a league-wide breakdown of optimal pitch selection rates. MLB players and organizations made the optimal pitch selection 23.1% of the time in 2025, and the non-optimal pitch selection 76.9% of the time. This paper does not necessarily make the claim that MLB pitch-calling was significantly poor more than three-quarters of the time, as the quantifiable effect of selecting the non-optimal pitches was negligible quite often, even when the optimal pitch technically was not selected.

Graph 2: League-wide optimal pitch selection rates



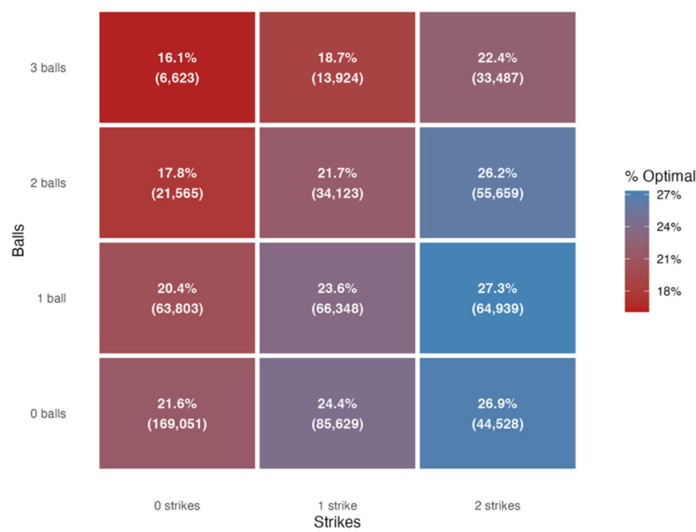
The following graph shows the average runs lost per pitch type over the 2025 MLB season. We can see that throwing the curveball (and similar variations, like the slurve or knuckle curve) had the least costly effects on run expectancy, indicating that a curveball may have been a "safer" pitch to throw. The cutter had the most costly effect on run expectancy, ignoring atypical pitches like the eephus.

Graph 3: Average runs lost by pitch type



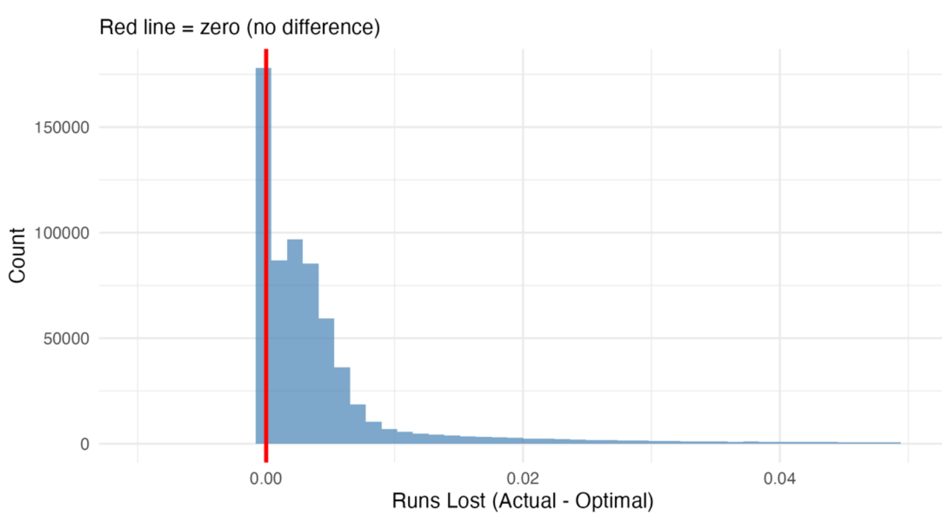
The following graph shows the optimal pitch rate by count. This examines the rates at which the ideal pitch was thrown in specific count states. Unsurprisingly, we see that non-optimal pitches are often thrown in counts that are advantageous to the batter, such as 3 balls and 0 strikes, while optimal pitches are often thrown in counts that are advantageous to the pitcher, such as 0 balls and 2 strikes. This aligns with conventional baseball logic, which says that a pitcher will have a greater advantage and more flexibility with their pitch selection in two-strike counts, whereas they have fewer options in three-ball and/or zero-strike counts because there is greater pressure to land a pitch in the strike zone.

Graph 4: Optimal pitch rate by count



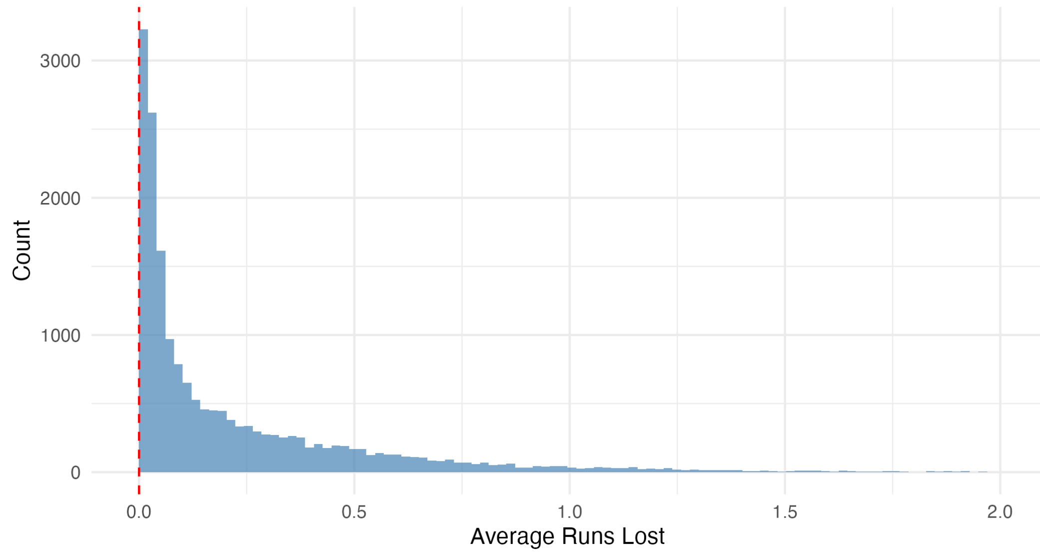
The following graph shows the distribution of runs lost per pitch. The distribution is highly right-skewed, with a large concentration of observations between 0 and 0.01 runs.

Graph 5: Distribution of runs lost per pitch



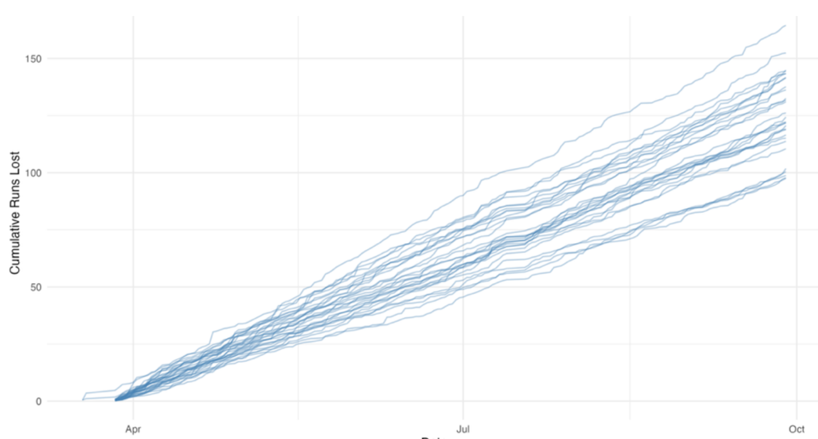
The following graph shows the average runs lost per individual pitcher appearance. Similar to the pitch-level results, the distribution is right-skewed, with most observations clustered near small positive values and a long tail extending toward higher run costs.

Graph 6: Average runs lost per pitcher appearance



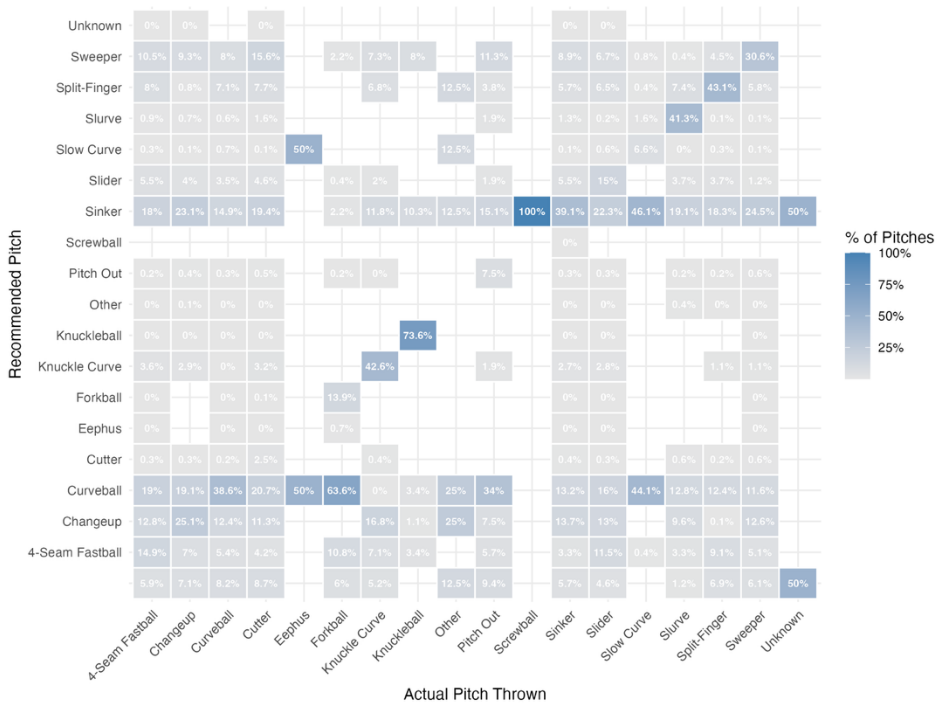
The following graph shows team-level cumulative runs lost totals over the full season. Each blue line represents one team, and the visualization tracks the trend of runs lost over the course of the season. Rates appear relatively constant, with no major league-wide blips at events such as Opening Day, the All-Star Break, the trade deadline, or the playoff push.

Graph 7: Team-level cumulative runs lost over full season



The following graph shows a breakdown of pitch type comparisons (recommended versus actual). We can see that the ideal counterfactual pitch for a corresponding actual pitch sometimes has a very different profile. For example, the most commonly recommended counterfactual to a cutter is not a similarly-moving slider, but rather a curveball with significantly more drastic negative induced vertical break. That is not always the case, though. The most commonly recommended counterfactual to a sinker is also a changeup; the two pitches have similar horizontal movement profiles, but different spin and velocity tendencies.

Graph 8: Actual versus recommended pitch type breakdown



Case Study: Miami Marlins

Given that this paper is heavily rooted in studying the effects of a decision to call pitches from the dugout, like the strategy that the Miami Marlins implemented, it becomes necessary to use this new modeling framework to analyze the Marlins' pitch-calling performance and success rate after they decided to shift the decision-making process to the dugout.

During the window in which the Marlins called pitches from the dugout (Sept. 19, 2025, through Sept. 28, 2025), the Marlins averaged fewer runs lost per pitch than the other 29 MLB teams. Compared to the average MLB team, the Marlins lost 0.0005 fewer runs per pitch, suggesting their strategy was more effective than the traditional method of allowing catchers to call pitches from behind home plate.

Table 10: Marlins vs. MLB average (Sept. 19-28, 2025)

Group	Pitches	Optimal %	Runs Lost	Per Pitch
Marlins	1445	0.218	9.03	0.0062
League Avg	41173	0.245	277.12	0.0067

Compared to the period before the Marlins shifted pitch-calling to the dugout (Opening Day through Sept. 18, 2025), the Marlins averaged fewer runs lost per pitch once they implemented the new dugout pitch-calling strategy (Sept. 19, 2025, through Sept. 28, 2025). The new strategy saved them an average of 0.0013 runs per pitch, suggesting their strategy was more effective than the traditional method of allowing catchers to call pitches from behind home plate.

Table 11: Marlins before vs. during implementation of dugout strategy

Period	Pitches	Optimal %	Runs Lost	Per Pitch
Before Sept 19	20580	0.223	154.07	0.0075
Sept 19-28	1445	0.218	9.03	0.0062

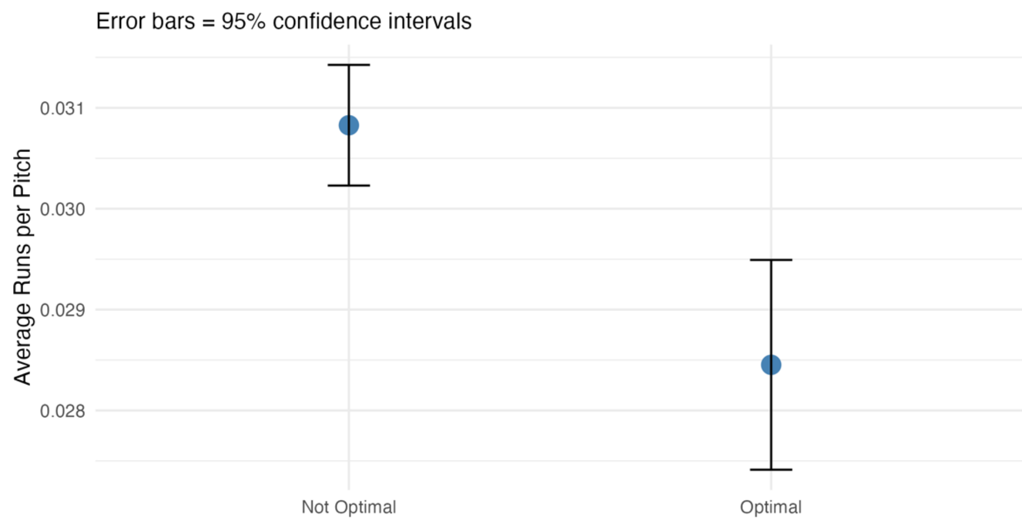
Statistical Validation

To evaluate whether suboptimal pitch selection decisions result in a measurable increase in run expectancy, a series of hypothesis tests was conducted on the runs lost metric.

A one-sample t-test was performed to determine whether the average runs lost per pitch was significantly greater than zero. The results indicate that the mean runs lost per pitch is 0.00696, which is significantly greater than zero ($t = 302.36$, $p < 0.001$). The 95% confidence interval for the mean runs lost is bounded below by 0.00692, further reinforcing the idea that making the optimal pitch selection results in fewer runs lost.

The following graph shows the average runs scored per pitch between optimal and non-optimal pitch selections, with error bars representing 95% confidence intervals. Pitches classified as non-optimal resulted in a higher average run value (0.0308) compared to optimal pitch selections (0.0285). This difference, while small on a per-pitch basis, is statistically significant and consistent with the idea that selecting the optimal pitch reduces run expectancy and benefits the defensive team.

Graph 9: Average runs scored on optimal and non-optimal pitches



To ensure that this result was not driven solely by pitch-level noise, the analysis was repeated at the pitcher-game level by aggregating runs lost across each pitcher's individual appearances. A one-sample t-test on this aggregated metric also yielded statistically significant results (mean = 0.2371 runs per appearance, $t = 97.03$, $p < 0.001$).

Finally, a two-sample Welch t-test was conducted to compare run outcomes between pitches on which the optimal pitch was selected and those on which it was not. The results show a statistically significant difference between the two groups ($t = 3.88$, $p < 0.001$). Non-optimal pitch selections resulted in a greater average of runs lost (0.03083) compared to optimal pitch selections (0.02845), with a 95% confidence interval for the difference ranging from 0.00118 to 0.00357.

The results of this series of hypothesis testing provide strong statistical evidence that deviations from the model's recommended pitch selection are not random, but rather are quite costly in terms of run expectancy. While the effect of any singular pitch may be small, the consistency and statistical significance of the results suggest that these costs accumulate meaningfully over a full season.

Discussion

This paper presents a framework for quantifying the run expectancy cost of pitch selection decisions in Major League Baseball. By training a LightGBM model on pitch-level Statcast data from 2021 through 2024 and applying it to the 2025 season, the study generates expected run values for empirical pitches and counterfactual alternatives, enabling a direct comparison between what was thrown and what the model identifies as optimal in any given situation.

The results reveal that pitch selection quality varies considerably across players and organizations. At the pitcher level, Jack Little of the Los Angeles Dodgers lost 0.0338 runs per pitch in 2025, ranking as the least efficient in the sample. Garrett Acton of the Tampa Bay Rays lost just 0.0004 runs per pitch in 2025, ranking as the most efficient in the sample. Anthony Seigler of the Milwaukee Brewers was least productive among catchers, losing 0.0165 runs per pitch, while Chad Wallach of the Los Angeles Angels performed best with just 0.0005 runs lost per pitch. The battery of Will Vest and Tomás Nido selected the optimal pitch on 41.67% of their pitches together, while the battery of Sam Long and Luke Maile lost 1.7551 runs over the season at 0.0325 runs per pitch. At the team level, the Chicago Cubs lost the fewest runs per pitch at 0.0059, while the Boston Red Sox lost the most at 0.0085. League-wide, the optimal pitch was selected just 23.1% of the time, underscoring the magnitude of the opportunity available to teams willing to invest in real-time pitch selection tools.

The Red Sox' net loss of 188.97 runs over a 162-game season — an average of more than one run per game — stands out as a glaring statistic. Organizations across the entire league (not just the Red Sox) must acknowledge that the current status quo of allowing catchers to call pitches in the traditional practice, rather than use advanced modeling processes to transmit pitch calls from the dugout in real-time, puts them at a competitive advantage. Given that modern implementations of the Pythagorean Expectation Formula in baseball attribute 10 runs to approximately 1 win, the Red Sox left approximately 18.897 wins on the table in 2025 — a difference between third place in the American League East (89-73) and first place in the American League East (108-54; also would have ranked first in all of Major League Baseball).

Several additional patterns emerged from the analysis. Pitch selection was most optimal in two-strike counts with fewer than three balls, and least optimal in hitter-friendly counts such as 3-0, consistent with conventional baseball logic suggesting pitchers face greater constraints on their pitch selection when behind in the count. At the pitch type level, the curveball and its variations generated the fewest average runs lost per pitch, while the cutter generated the highest among standard pitch types. The recommended counterfactual pitch type was frequently a pitch with a meaningfully different movement profile than what was actually thrown, suggesting that the model withstood any biases surrounding movement patterns.

Taken together, these findings build on prior research by Bock (2015), Prasad (2021), and Melville (2023) by introducing a directly actionable, outcome-based evaluation framework. Rather than simply identifying patterns in how pitches are selected, this study places a quantifiable run cost on each deviation from the model's recommendation, producing leaderboards and visualizations that organizations could use to evaluate personnel and inform in-game strategy.

The most significant practical implication of this research is the potential for MLB organizations to transmit model-generated pitch calls from the dugout to the field in real time. Given that the model input can be summarized from basic data of the current game state and the current pitcher's arsenal, teams (while complying with MLB technology policies) could introduce a system to generate an ideal pitch call for the upcoming pitch at the exact instant the previous pitch/event concludes.

As demonstrated by the Miami Marlins during the 2025 season, pitch call transmission from the dugout is both permitted under MLB rules and operationally feasible, and this paper shows that the Marlins' strategy is also capable of having a positive, run-saving effect. This paper provides an analytical foundation for an overhauled pitch-calling process, suggesting that organizations willing to adopt data-driven pitch calling could recover a meaningful number of runs (and, therefore, wins) that would otherwise be lost to suboptimal decisions made under time pressure and without access to real-time predictive tools.

Future research could extend this framework in several directions. Incorporating pitch location, handedness platoon, and tunneling effects, as noted by Melville, would improve the counterfactual generation process. Expanding the model to account for batter-specific tendencies could improve prediction accuracy, and making adjustments for the effects of park factors, weather, and pitcher fatigue could also improve the model, predictions, and recommendations. Additionally, incorporating a rolling window of pitch-level metrics (rather than applying a full-season average) would allow the framework to hold stronger despite potential midseason mechanical tweaks. Finally, examining whether the run cost of suboptimal pitch selection is greater for lower-leverage or lower-Stuff pitchers could allow these organizations to make player-specific strategy decisions that maximize run and win expectancy.